

Assignment 1

Complete the 2 strategy classes MSCSAIChecker and MSCSCyberChecker.

The Strategies all implement the interface Checker.

The basic Strategy MSCSChecker is provided. It shows how to read the text file “elecs.txt” which includes the current list of MSCS elective courses and store it in a list.

- The files “elecsAI.txt” and “elecsCyber.txt” are provided.
- The driver is MSCSGradTester, which does compile and run on Java 22 and later, the file Checker.java explains what to change if your Java is older.

In MSCSChecker, the method satisfies checks that verify:

https://bulletin.binghamton.edu/preview_program.php?catoid=7&poid=2210&returnto=303

- Please note that a Set is used to declare “valid” to remove courses that are repeated.

Later on in the same page, you find the rules for the AI track and for the Cybersecurity Track.

- Please note that CS 536 and CS 537 are mentioned in the Cybersecurity Track but that could reasonably be expressed as “one of CS336, CS537, and CS580E can be used in the Cybersecurity Track” because there are still students in the Track that took CS580E before it was renumbered as CS537.

I am asking you complete the MS requirements, because the BS program has many additional checks and other options in the electives.

The output shows that more checks are needed for 4+1 students because of conflicts between meeting requirements in the undergraduate and graduate tracks.

The “satisfies” check is only about courses taken, it does not check the GPA requirement, graduate residence requirements, or the completion of the project or thesis.

Additional details are in the Graduate Student Handbook, there will be a link to that on Monday.